ÜSKÜDAR UNIVERSITY

FACULTY OF ENGINEERING AND NATURAL SCIENCES

SOFTWARE ENGINEERING

DATA SCIENCE AND ANALYTICS

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Course Code	IE 333
Project Assignment	Exploratory Data Analysis (EDA) and Classical Modeling for Time Series Data
Group Number	10

Table of Contents for Project Report

Chapters	Description	Marks
Chapter 1: Introduction	**1.1 Problem Statement** • Define the objective (e.g., "Forecast monthly sales using historical data"). **1.2 Dataset Overview** • Source, variables, time range, and relevance. **1.3 Key Questions to Answer** • Example: "Does the data show seasonality? Which model performs best?"	• **Problem Statement (5 Marks):** Clear definition of objectives and relevance • **Dataset Overview (5 Marks):** Description of source, variables, and time period
Chapter 2: Data Collection & Preprocessing	**2.1 Data Loading** • Code snippet to load data (e.g., pd.read_csv()). **2.2 Data Cleaning** • Handling missing values, outliers, duplicates. **2.3 Feature Engineering** • Create time-based features (e.g., Month, Day_of_Week). **2.4 Data Splitting** • Train-test split (e.g., 80-20) for model evaluation.	• **Data Loading (5 Marks)**: Proper import and initial inspection • **Data Cleaning (10 Marks)**: Handling missing values, outliers, and duplicates •
Chapter 3: Exploratory Data Analysis (EDA)	**3.1 Descriptive Statistics** • Mean, median, variance, skewness. **3.2 Time Series Decomposition** • Trend, seasonality, and residual analysis. **3.3 Visualizations** *** Line Plot** (Trend over time). *** Histogram/Box Plot** (Distribution of target variable). *** Correlation Heatmap** (Feature relationships).	• **Descriptive Statistics (5 Marks)**: Mean, variance, skewness, etc. • **Time Series Decomposition (5 Marks)**: Trend, seasonality, residuals • **Visualizations (5 Marks)**:
Chapter 4: Modeling	**4.1 Model 1: ARIMA** • **Model Explanation** (p, d, q parameters). • **Implementation** (Code + hyperparameter tuning). • **Forecast Visualization** (Plot predictions vs. actual). **4.2 Model 2: Regression-Based** • **Linear Regression** (Trend modeling). • **Polynomial Regression** (Nonlinear trends). • **Feature Importance** (Coefficient analysis). **4.3 Model 3: Fourier Series** Detect seasonality frequencies.	• **Model 1 Implementation (10 Marks)**: ARIMA (code + hyperparameter tuning) • **Model 2 Implementation (10 Marks)**: Regression (linear/polynomial/forier/etc)
Chapter 5: Model Evaluation & Comparison	**5.1 Evaluation Metrics** • **RMSE** (Root Mean Squared Error). • **MAPE** (Mean Absolute Percentage Error). • **R^2** (R-squared). Example of evaluation results: **5.2 Results Table** Model | RMSE | MAPE | R^2 ARIMA(1,1,1) | 120.5 | 8.2% | 0.89 Polynomial Reg | 95.3 | 6.7% | 0.92 **5.3 Discussion** ◦ Which model performed better? Why? ◦ Limitations of each approach.	• **Metrics Calculation (5 Marks)**: RMSE, MAPE, R^2 for both models • **Results Comparison (5 Marks)**: Table + critical discussion of performance
Chapter 6: Conclusion & Recommendations	**6.1 Key Findings** • Summarize trends, seasonality, and model performance. **6.2 Business/Research Implications** • How can results drive decisions? **6.3 Future Work** • Suggestions for improvement (e.g., try SARIMA, add external variables).	**5 marks**
Submission files (via STIX)	1. .ipynb python file 2. Dataset (Excel or CSV) 3. Jupyter notebook exported as PDF 4. PDF report	
Submission files (face to face)	1. Printed copy of the report	The project table of content must be included in the report (i.e. this table) for marking the project.
Presentation	Presentation schedule will be provided later on.	**25 marks**

Chapter 1: Introduction

1.1 Problem Statement:

Objective: The primary objective of this project is to analyze and forecast the monthly number of international airline passengers using historical data. We aim to identify underlying patterns like trend and seasonality and build predictive models to estimate future passenger traffic.
Relevance: Accurate forecasting of airline passengers is crucial for the aviation industry. It aids airlines in resource allocation (aircraft, crew scheduling, fuel planning), airports in managing operations (staffing, gate assignments, facility management), and related businesses (hotels, tourism) in planning and investment decisions. Understanding passenger volume trends helps in strategic planning and optimizing revenue.

1.2 Dataset Overview:

Source: The dataset used is the classic "AirPassengers.csv" dataset, widely used for time series analysis examples. It contains monthly totals of international airline passengers from 1949 to 1960. https://www.kaggle.com/datasets/rakannimer/air-passengers
Variables:
Month: The month and year of the observation (used as the time index).
#Passengers: The number of international airline passengers for that month (the target variable).
Time Range: The data spans from January 1949 to December 1960, covering 12 years (144 observations).Relevance: This dataset is highly relevant for demonstrating time series forecasting techniques due to its clear trend and seasonal components.

1.3 Key Questions to Answer:

This analysis seeks to answer the following key questions:
1. What is the overall trend in airline passenger numbers over the observed period?
2. Does the data exhibit seasonality? If so, what is the pattern?
3. Can we build time series models (like ARIMA) and regression models to forecast future passenger numbers?
4. Which implemented model provides the most accurate forecasts based on standard evaluation metrics?

Chapter 2: Data Collection & Preprocessing

2.1 Data Loading:

The dataset was loaded into a pandas DataFrame using the read_csv function. The 'Month' column was converted to datetime objects, which were then set as the DataFrame index to facilitate time series analysis.

```
import pandas as pd

df = pd.read_csv("AirPassengers.csv")
df['Month'] = pd.to_datetime(df['Month'])
df.set_index('Month', inplace=True)
```

2.2 Data Cleaning:

The dataset was checked for potential data quality issues:

- **Missing Values**: A check using df.isnull().sum() was performed.
  Output from isnull().sum()
  Passengers 0
  No missing values were found in the dataset.
- **Duplicates**: A check for duplicate rows using df.duplicated().sum() was performed.
  # Output from duplicated().sum()
  # 0
- **Outliers**:

2.3 Feature Engineering:

To aid analysis and modeling, especially for regression-based approaches, time-based features were extracted from the datetime index:
Month_num: The numerical month (1-12) to potentially capture seasonality.
Year: The year of the observation to potentially capture the trend.
TimeIndex: A simple numerical sequence (0, 1, 2, ...) representing the time steps, primarily used for modeling the trend in regression models.

```
df['Month_num'] = df.index.month
df['Year'] = df.index.year
df['TimeIndex'] = np.arange(len(df))
```

2.4 Data Splitting:

For model evaluation, the data was split into a training set and a test set. An 80-20 split is common, but given the time series nature and the desire to forecast the last year, the data up to the end of 1959 was used for training, and the data for 1960 was held out as the test set.
Training Set: January 1949 - December 1959
Test Set: January 1960 - December 1960

```
# Code Snippet for Data Splitting
train = df[:'1959']
test = df['1960':]
```

Chapter 3: Explatory Data Analysis (EDA)

3.1 Descriptive Statistics:

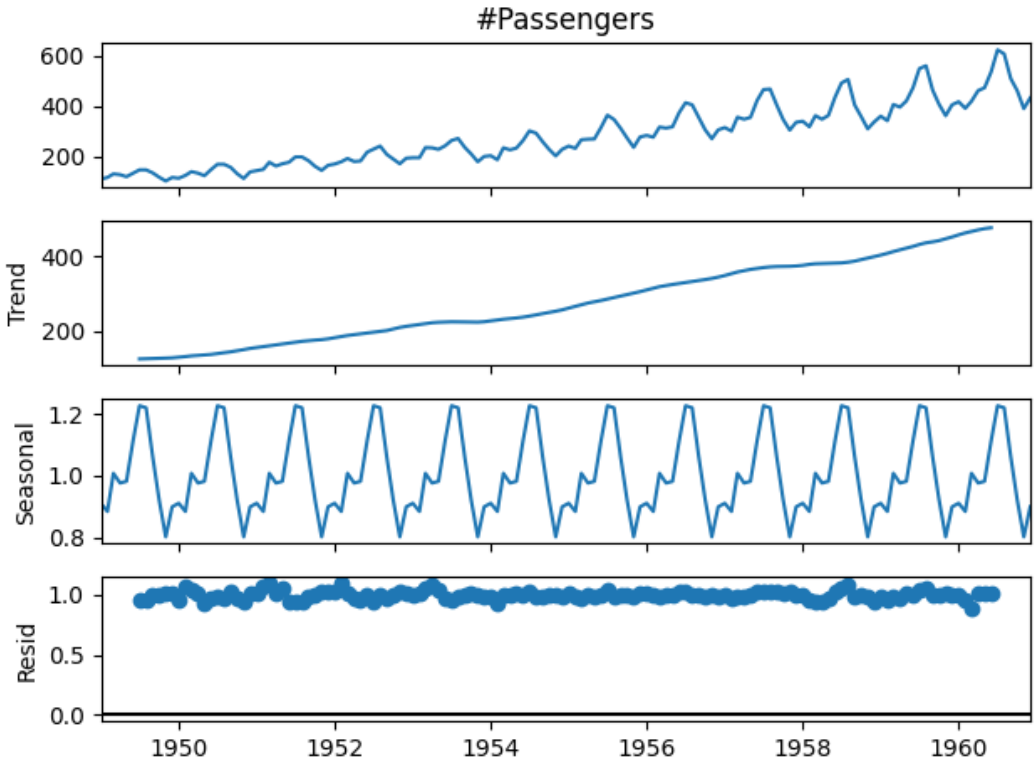
The dataset was loaded into a pandas DataFrame using the read_csv function. The 'Month' column was converted to datetime objects, which were then set as the DataFrame index to facilitate time series analysis.

```
print("Descriptive Stats:\n")
print("Mean:", df['#Passengers'].mean())
print("Median:", df['#Passengers'].median())
print("Variance:", df['#Passengers'].var())
print("Skewness:", df['#Passengers'].skew())
```

Mean: The average number of passengers is 280.3, indicating the central tendency of the data.
Median: The middle value is 265.5, slightly lower than the mean, suggesting some influence from higher passenger counts.
Variance: With a variance of 14,391.92, the data shows considerable spread, meaning passenger numbers fluctuate significantly.
Skewness: The positive skewness (0.58) confirms a right-skewed distribution, where a few unusually high passenger counts pull the mean above the median.

3.2 Time Series Decomposition:

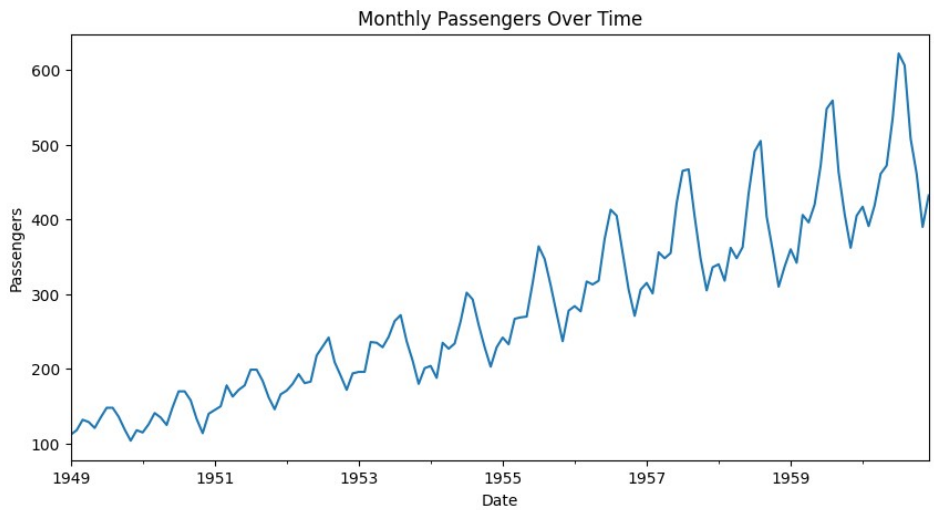
Seasonal decomposition helps separate the time series into its core components: Trend, Seasonality, and Residuals. A multiplicative model was chosen because the seasonal fluctuations appear to grow proportionally with the trend.



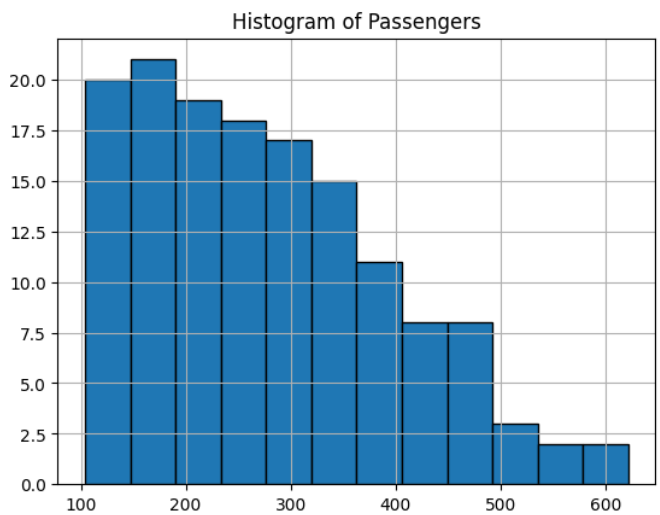
Trend: The decomposition clearly shows a strong upward trend throughout the period, indicating consistent growth in air travel.
Seasonality: A distinct yearly pattern is visible, with peaks typically occurring in the summer months (July/August) and troughs in the winter months (November/December).
Residuals: The residuals appear relatively random around 1.0, suggesting the multiplicative model captured the main trend and seasonal components reasonably well, although some patterns might remain.

3.3 Visualizations:

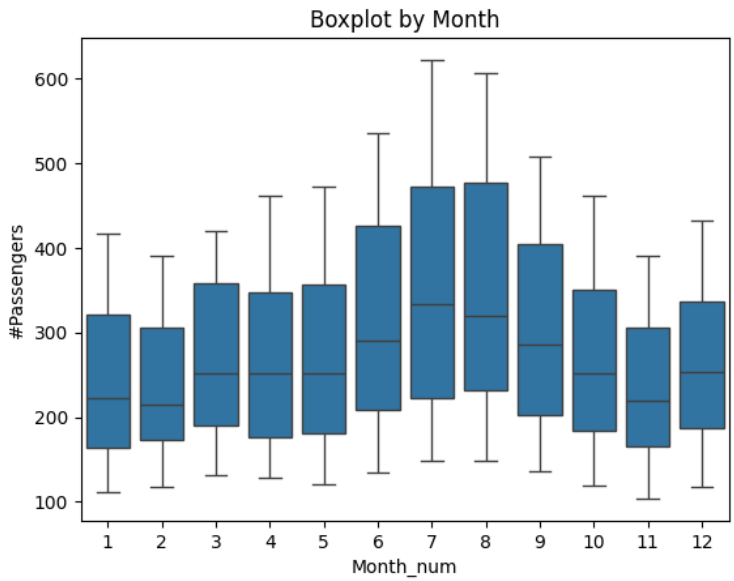
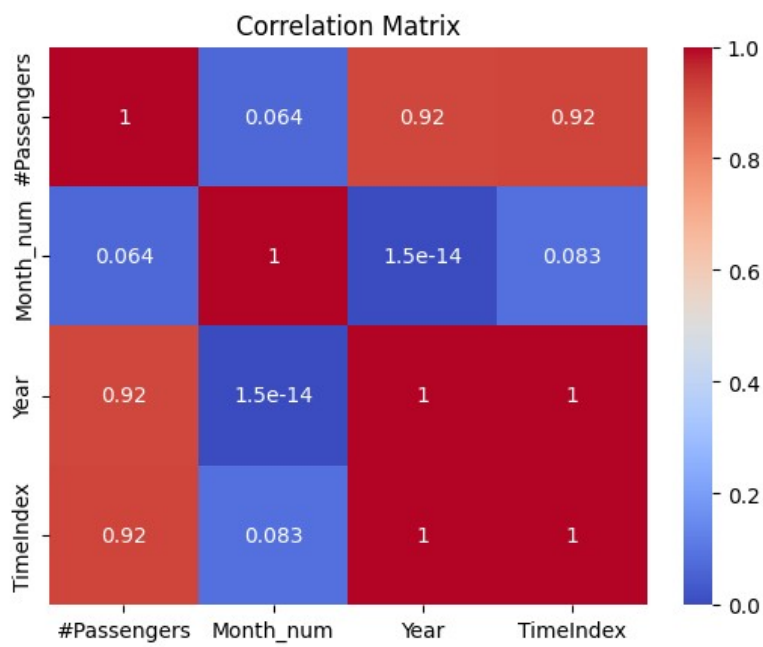
Line Plot (Trend over time): This plot visually confirms the strong upward trend and the recurring seasonal pattern identified in the decomposition.



Histogram (Distribution of target variable): The histogram shows the distribution of monthly passenger counts. It's unimodal but spread out and slightly right-skewed, consistent with the descriptive statistics.



Box Plot (by Month): This plot provides a clearer view of the seasonality. It shows the distribution of passenger counts for each month across the years. Higher median values and wider ranges are typically seen in months 6, 7, and 8 (June-August), while lower values occur around months 11 and 12 (November-December).



Correlation Heatmap: The heatmap shows the linear correlation between the variables.